

Development of an IoT and Mobile App-based Real-time Remote e-health Care System for Rural Communities in Bangladesh

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Abstract— This work presents an IoT-based remote health monitoring system for rural communities in Bangladesh, aimed at improving healthcare access, particularly in remote areas. The system employs sensors such as LM35, AD8232, and SEN-11574 to measure vital health parameters, including body temperature, ECG and heart (pulse) rate. The collected data is transmitted to an IoT cloud for recording and analysis. The system consists of three primary components: patient condition detection, data storage on the cloud, and remote data viewing. This setup enables healthcare providers to monitor patients' health remotely, reducing the need for frequent hospital visits. Designed as a non-invasive, cost-effective solution using locally available sensors, it is easy to implement and highly accessible. The stored cloud data allows for longitudinal health analysis, offering valuable insights into patients' conditions. By leveraging IoT technology, this system ensures continuous patient monitoring and the prompt detection of health abnormalities. Its affordability and practicality make it a transformative solution for enhancing healthcare delivery in rural areas of Bangladesh.

Keywords— *IoT, e-health care, Mobile apps, Rural communities, Microcontroller.*

I. INTRODUCTION

In rural areas of Bangladesh, accessing professional healthcare services remains challenging due to distance barriers and a shortage of doctors [1-2]. One effective solution to this issue is the implementation of a remote patient monitoring system. This system utilizes IoT devices, which are increasingly prevalent and impactful in healthcare. By enabling early detection of health issues, IoT devices can help reduce medical costs and address the growing concerns about health conditions [3-4]. Over the years, there has been a heightened awareness of health issues, necessitating continuous monitoring of individuals' health. The evolution of health monitoring systems has been rapid, offering the potential to revolutionize healthcare delivery. However, the effectiveness of smart health monitoring systems in clinical settings is still a topic of debate. Smart healthcare is

particularly crucial for patients requiring ongoing monitoring, especially in rural areas where access to hospitals may be limited [5]. These systems facilitate communication between rural clinics and city hospitals, ensuring timely intervention and coordinated care for patients. Consequently, healthcare monitoring systems have seen a significant increase in adoption across hospitals and healthcare facilities [6].

IoT-based remote health monitoring systems have been developed to improve the quality and accessibility of healthcare services. These systems utilize sensors to measure vital signs and collect data from patients, which is then transmitted wirelessly to a central monitoring station for remote viewing [7-9]. The use of IoT in healthcare allows for constant monitoring of patients, enabling early detection and intervention of potential health problems [10]. These systems can track health metrics such as body temperature, blood pressure, and heart rate, providing personalized medical history and real-time updates on physiological parameters [11]. Over 60 percent of people in Bangladesh live in rural areas, where medical services are insufficient, creating a serious problem [12]. Rural people in Bangladesh face common diseases such as diarrhea, malaria, malnutrition, and more due to the lack of a proper healthcare system [13-14]. Because of the lack of properly trained healthcare providers, these people often travel to urban areas for treatment, which is not feasible for everyone in rural areas [15]. Digital health is the practice of quickly delivering health services using technology, such as computers, smartphones, and internet [16].

In our proposed e-healthcare system for rural communities in Bangladesh, we have integrated IoT devices to monitor vital health data such as pulse (heart rate), ECG, and body temperature of patients. Additionally, doctors can monitor patients' data remotely, anytime and anywhere, through a mobile app. This system has the potential to enhance quick access to healthcare, reduce wait times, offer remote consultations, and deliver health education. It can significantly help bridge the healthcare gap between rural and urban areas.

The rest of the paper is organized as follows: Section II discusses related works with a critical analysis. Section III describes key components of the proposed system. Section IV provides a detailed explanation of the proposed e-healthcare system. Section V presents the results and analysis. Finally, Section VI concludes the paper with key findings.

II. LITERATURE REVIEW

The integration of IoT-based health monitoring systems has revolutionized healthcare by enabling seamless connectivity between devices to gather and transmit patient health data in real time. These systems leverage advanced sensors to monitor vital parameters such as heart rate, pulse rate, blood pressure, ECG and body temperature etc., transmitting this critical information to healthcare providers over the Internet. Such technology facilitates remote patient monitoring, empowering doctors to track patient health from a distance and intervene promptly when abnormalities are detected. By analyzing the collected data, healthcare professionals can identify trends, anticipate potential health issues, and implement timely interventions, thus enhancing preventive care and promoting overall well-being. This approach holds significant potential for addressing the healthcare challenges faced by rural communities, where access to medical facilities and resources is often limited. Many researchers are actively conducting studies to develop such systems. Banerjee et al. [17] developed an IoT-based healthcare system to monitor basic health parameters, such as body temperature and pulse rate. The system utilized the ESP32 microcontroller in conjunction with ECG, pulse, and temperature sensors. The ESP32 microcontroller collected data from the sensors, processed it, and transmitted it to the IoT cloud. However, a significant limitation of this system was its lack of integration with a mobile application and the absence of more advanced health-monitoring sensors. Another system described in [18] was developed to calculate the pulse rate using a mobile phone's LED and camera instead of traditional sensors. The phone's light and camera were used to track blood volume and calculate the pulse rate. The calculated pulse rate was then transmitted wirelessly to a computer. While this system offers a convenient method for pulse rate calculation, it has limitations in scenarios requiring continuous monitoring, as it cannot provide ongoing tracking. Trivedi et al. [19] proposed a system based on Arduino for monitoring basic health parameters. The system uses an Arduino Uno microcontroller and transmits data via a Bluetooth module. However, a limitation of this system is the small coverage area of the Bluetooth module, which makes it unsuitable for transmitting data over long distances. Kumar et al. [20] and Hossain et al. [21] proposed systems based on Arduino that use a Wi-Fi module to ensure compatibility with the IoT cloud, allowing data to be transmitted anywhere. These systems utilize a temperature sensor (DS18B20), a pulse sensor, and an Arduino Uno with a Wi-Fi module. The ThingSpeak server is used as the IoT cloud platform to receive the data, which is then visualized in various formats such as charts, histograms, and normal values. Although these systems feature a solid architecture, they are limited by the use of only a few sensors and the lack of software for data processing and analysis.

The contribution of our work lies in the integration of pulse (heart) rate, body temperature, and ECG sensors to enhance the remote patient monitoring system. We used the ESP32 microcontroller, which is integrated with a Wi-Fi module, offering a low-cost and simplified alternative to the Arduino

Uno. Additionally, we employed ThingSpeak for data visualization, retrieving data from the ThingSpeak channel and processing it according to the required level. MATLAB was utilized for signal processing, particularly for ECG signal analysis and for measuring pulse (heart) rate through ThingSpeak visualization. Furthermore, we interfaced an Android app with ThingSpeak, enabling convenient health data monitoring from any location.

III. KEY COMPONENTS

A. ESP32 Microcontroller

The ESP32 is an advanced and versatile microcontroller. It features built-in Wi-Fi (802.11) and dual-band Bluetooth, allowing easy wireless connectivity with other devices and servers. This makes it a popular choice for Internet of Things (IoT) applications. The ESP32 includes numerous peripheral pins, such as GPIO pins, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), and more, enabling seamless connection with both analog and digital sensors. It is developed using a dual-core microprocessor and a 40nm process, ensuring high performance. The microcontroller comes with 4 MB of flash memory and an additional 8 MB of pseudo-static RAM (PSRAM). For security, it supports IEEE 802.11 standard security features. The ESP32 can execute various programming languages, including C/C++ and MicroPython, accommodating a wide range of developers. [22], is shown in Fig. 1. Its low power consumption makes it ideal for IoT-based projects where power efficiency and compact size are critical.



Fig.1. ESP32 Microcontroller

B. Heartbeat Sensor

A heartbeat sensor measures pulse rate by detecting changes in the blood volume of arteries as the heart pumps. The sensor includes a Light Emitting Diode (LED) and works based on reflected light. When the sensor is placed on a fingertip and the heart pumps, the blood volume in the arteries increases, causing more light to be absorbed and reflected back. The amount of reflected light depends on the blood volume. When the blood volume increases, more light is absorbed; when it decreases, less light is absorbed. This variation generates an electrical signal, which is then processed into pulse rate data using an Arduino program. The pulse sensor has three pins: Vcc, ground, and signal. The heartbeat sensor is low-cost and compatible with the ESP32 microcontroller and other microcontrollers. It is manufactured by a renowned electronics company, is compact in size, and can be integrated into wearable devices. Some properties of the sensor are:

- Created for plug and play
- Operating voltage is 3v to 5v

- Works with esp32 with an ADC

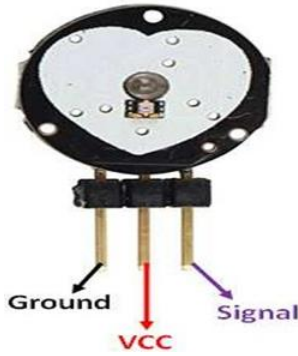


Fig. 2. Heartbeat sensor (SN-11574)

C. Electrocardiograph ECG Sensor

The AD8232 sensor is an integrated circuit designed for electrocardiogram (ECG) signal conditioning. It features a 3-electrode configuration, which is placed on the human body to extract ECG signals. This sensor includes several special features, such as a 2-pole high-pass filter, a 3-pole low-pass filter, high signal gain, and a common-mode rejection ratio (80 dB). These features ensure the extraction of low-level ECG signals from the body, along with proper filtering and amplification for accurate ECG signal conditioning. The operating voltage of the sensor ranges from 2.7 to 6 volts, with a supply current of 750 μ A. Its output current is high, typically $\pm 250 \mu$ A. The AD8232 sensor is low-cost, compact, portable, and easily compatible with the ESP32 microcontroller [23].



Fig.3 ECG sensor (AD8232)

D. Body Temperature Sensor

The LM35 temperature sensor detects temperature and calculates it in degrees Celsius using its analog output voltage. The output voltage is linearly proportional to the temperature in Celsius. The sensor has a gain of 10 mV per degree Celsius, meaning a voltage of 140 mV corresponds to 14 degrees Celsius. This sensor is simple, low-cost, and easy to use. It has three pins: power, ground, and output [24]. The temperature sensor illustrated in Fig.4. The characteristics of the sensor are as follows:

- The power supply of the sensor is 4v to 20v
- The temperature range of the sensor is -55°C to 155°C
- Accuracy of the sensor is $\pm 0.5^\circ\text{C}$.

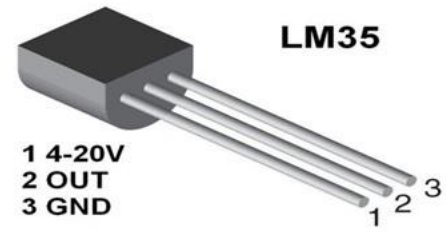


Fig. 4. Body temperature sensor (LM35)

E. IoT Platform

ThingSpeak is an IoT cloud platform and open-source software written in Ruby. It has numerous library files and is integrated with Math Works and MATLAB. ThingSpeak receives data from IoT devices using API keys over Wi-Fi internet from anywhere. A read application programming interface (API) key is used to read sensor values, which are then stored and visualized in a ThingSpeak channel. A write API key is used to write data into a channel on ThingSpeak. This data can be visualized and analyzed using MATLAB. ThingSpeak allows real-time monitoring and stores data that can be exported to Excel. It can also connect with custom mobile applications. However, the free version of ThingSpeak has limitations, while the paid version supports more advanced applications [25].

F. Thingview Mobile App

ThingView is the perfect tool to view and interact with our ThingSpeak channels in an easy way. It works for both public and private channels. With it, we can view data in a variety of chart types and have the possibility to set the time frame, the date range, and even the number of results to show. We can also download and share data with our friends. To use ThingView, we need to have a ThingSpeak account and a valid API key. The application itself is very simple and easy to use. Simply enter the channel ID to get the data for that channel. The application will automatically display the data in the form of a chart.

IV. PROPOSED E-HEALTH CARE SYSTEM

The proposed system is an IoT-based health parameter monitoring system designed to remotely track a patient's vital signs using an ESP32 microcontroller and three health sensors: a body temperature sensor, a pulse sensor, and an ECG sensor. The LM35 temperature sensor is used to measure body temperature from the patient's fingertip, while the pulse sensor (SEN-11574) detects changes in blood volume within the arteries to calculate the pulse rate in beats per minute (BPM). Additionally, the ECG sensor (AD8232) records the heart's electrical activity, providing real-time ECG signals. The ESP32 microcontroller serves as the system's core, managing sensor data acquisition, processing, and transmission. With built-in WiFi and Bluetooth capabilities, the ESP32 seamlessly transfers data to an IoT cloud platform, enabling remote health monitoring.

The system utilizes ThingSpeak, an IoT cloud platform that collects, stores, visualizes, and analyzes sensor data. ThingSpeak supports MATLAB integration, allowing advanced data processing and analysis. It provides various visualization tools, such as graphs, charts, gauges, and

numerical widgets, making health monitoring more interactive. Each patient is assigned a unique channel ID and API key, ensuring secure access to their health records. Doctors can remotely review patient data at any time, analyze historical trends, and make informed medical decisions. The platform offers two types of channels: public and private. Private channels restrict access through an API key, ensuring data confidentiality. In the free version, data updates occur every 15 seconds, whereas the paid version supports updates every 1 second for higher data accuracy.

The system also integrates with a mobile application, allowing users to monitor health parameters conveniently. The app retrieves real-time data from the IoT cloud, providing instant access to patient vitals. The overall workflow begins with sensor initialization, followed by continuous health data acquisition. The ESP32 processes and transmits the collected data to ThingSpeak, where it is stored, visualized, and analyzed. Doctors and caregivers can access patient data remotely via the IoT platform or mobile application, ensuring timely medical intervention when necessary. This IoT-based remote health monitoring system is cost-effective, efficient, and portable, making it ideal for real-time patient care. By leveraging wireless connectivity and cloud-based data storage, it eliminates geographical barriers, enabling doctors to monitor patients from any location. The system's ability to store historical data in CSV format further enhances medical record-keeping and long-term health tracking. The block diagram of the proposed system is illustrated in Fig. 5, while the circuit diagram, designed using Fritzing software, is shown in Fig. 6. The flowchart in Fig. 7 outlines the entire process, from sensor initialization and data transmission to cloud storage and real-time visualization. This system provides a smart and effective healthcare solution, addressing the growing need for remote patient monitoring through IoT technology.

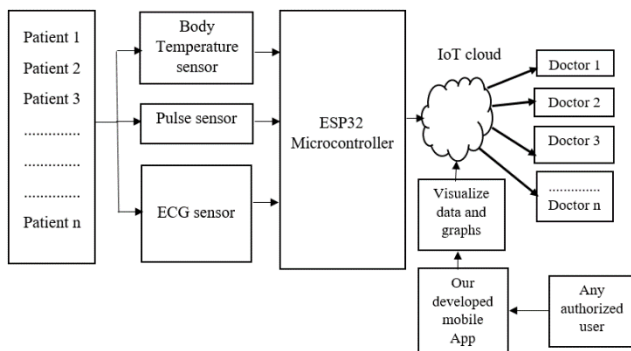


Fig. 5 Block diagram of the proposed e-health care system.

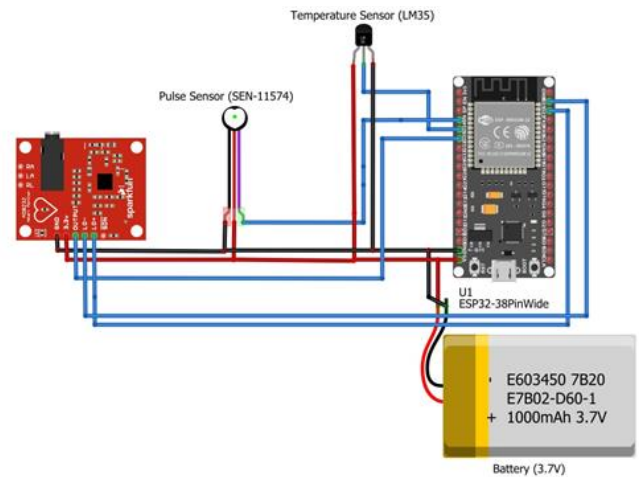


Fig. 6. Schematic circuit diagram of the e-health care system.

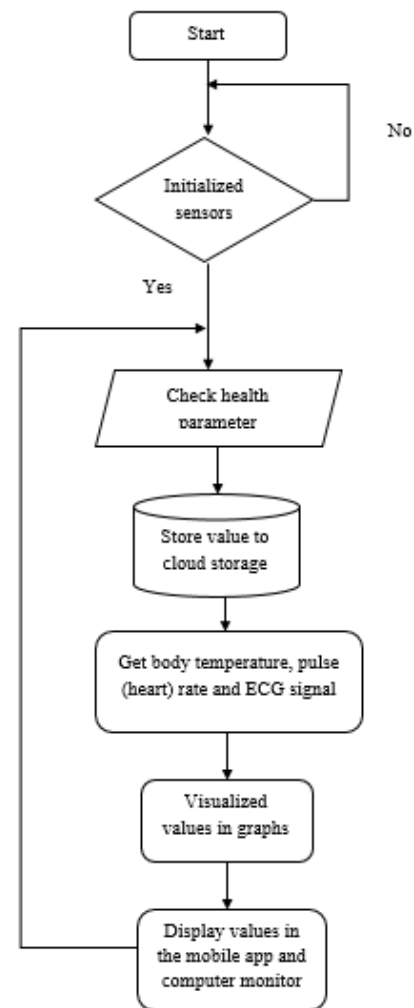


Fig.7. Flow chart of the e-health care system.

The prototype of the system is shown in Fig. 8. All the sensors, including the ECG sensor (AD8232), pulse sensor (SEN-11574), and body temperature sensor (LM35), are connected to the ESP32 microcontroller. The ESP32 collects data from these sensors, processes it, and transmits it to the ThingSpeak IoT cloud via WiFi over the internet. The system interfaces with both the patient and doctor through a smartphone application, which retrieves real-time health data from the ThingSpeak platform. This seamless integration

enables remote health monitoring, allowing doctors to access patient data anytime, ensuring timely medical intervention.

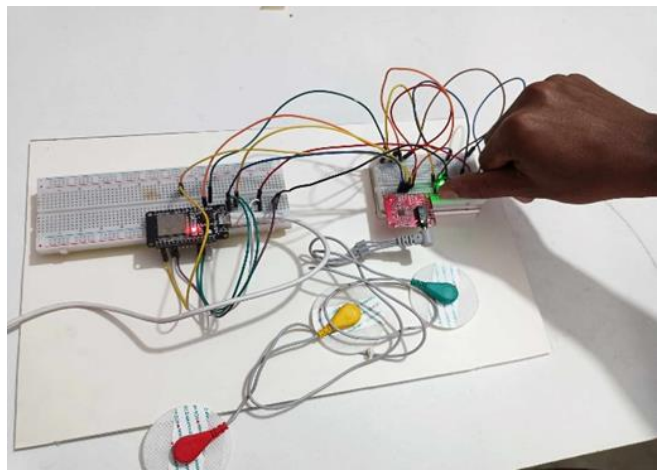


Fig.8. Prototype of the proposed e-health care system.

V. RESULTS AND ANALYSIS

Table 1 presents the standard health parameters for a healthy individual under normal conditions. Fig. 9 illustrates the pulse rate measurements in beats per minute (BPM), collected using the proposed system. The raw pulse rate data is processed in MATLAB to compute the BPM values. After measuring the pulse rate, the system transmits the data to the ThingSpeak server using an API key. Fig. 10 visualizes the recorded pulse rate data on the ThingSpeak platform. From the Fig, 10, it is observed that the measured pulse rate from the developed prototype falls within the range of 92–100 BPM, which is considered normal. Fig. 11 demonstrates the experimental setup for ECG measurement, while Fig. 12 displays the ECG signal obtained from the designed system. The collected ECG data undergoes signal filtering using MATLAB to remove noise and enhance signal clarity. After processing, the filtered ECG data is stored in a CSV file and subsequently uploaded to the ThingSpeak server. The visualization of the ECG data on ThingSpeak is presented in Fig. 13. Finally, Fig. 14 presents the fingertip (body) temperature measurement results of a patient, processed in MATLAB, using data collected from our developed system. Fig. 15 displays the corresponding temperature data as visualized on the ThingSpeak platform.

TABLE 1 HEALTH PARAMETERS UNDER NORMAL CONDITION

Health parameter	Normal condition
Pulse (heart) rate	60-120 BPM
Body Temperature	36.2-37.8° C

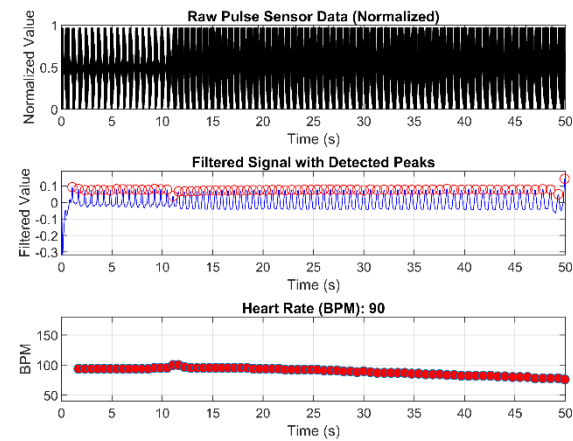


Fig. 9. Pulse rate measurement in MATLAB software

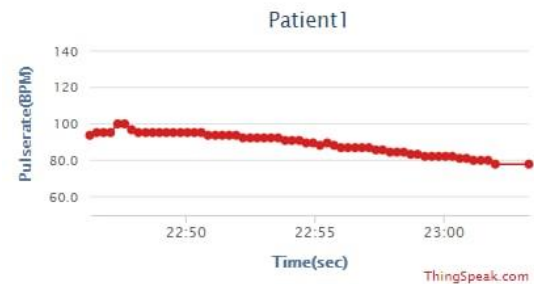


Fig.10. Pulse rate in ThingSpeak.



Fig.11. ECG measurement setup.

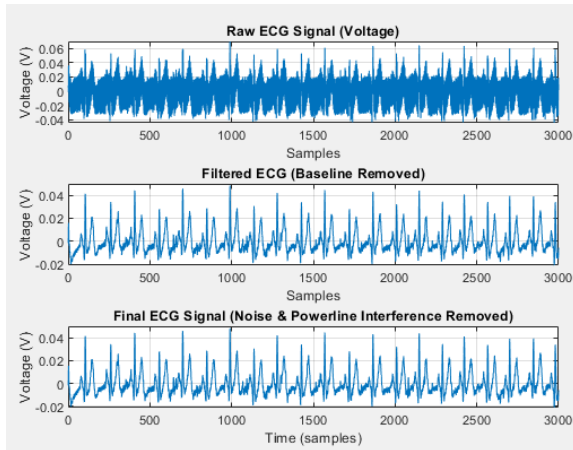


Fig. 12. Measured ECG signal in MATLAB.

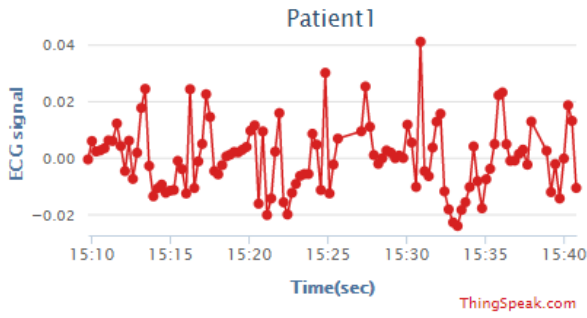


Fig.13 Measured ECG signal in ThingSpeak.

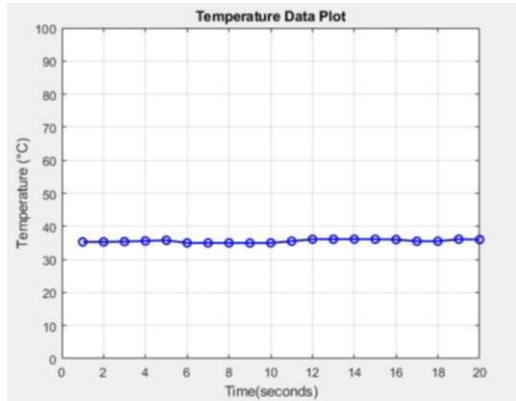


Fig.14. Measured fingertip (body) temperature in MATLAB.

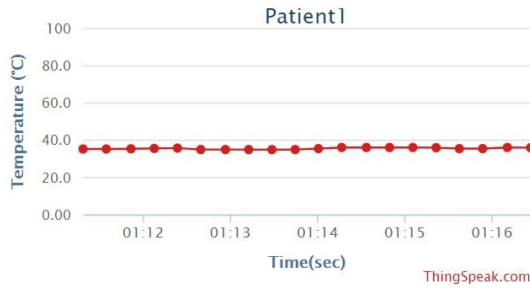


Fig.15. Measured fingertip (body) temperature in ThingSpeak.

An Android app, ThingView, has been integrated with our proposed system, as shown in Fig. 16, where three doctors are considered; however, the system is scalable, allowing more doctors to be added as needed. This integration enables doctors to remotely access real-time health data from any location at any time, ensuring continuous monitoring and timely medical intervention. Fig. 17 presents the ThingView interface for Doctor-1, where they can monitor all medical data related to Patient-1. Similarly, multiple patients can benefit from this e-healthcare system, and any authorized doctor or healthcare professional can securely access the health data of their assigned patients. This ensures an efficient, scalable, and accessible remote healthcare monitoring solution.

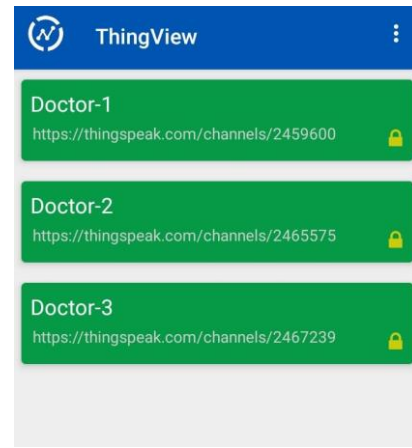


Fig. 16. Doctors' main window at ThingView.

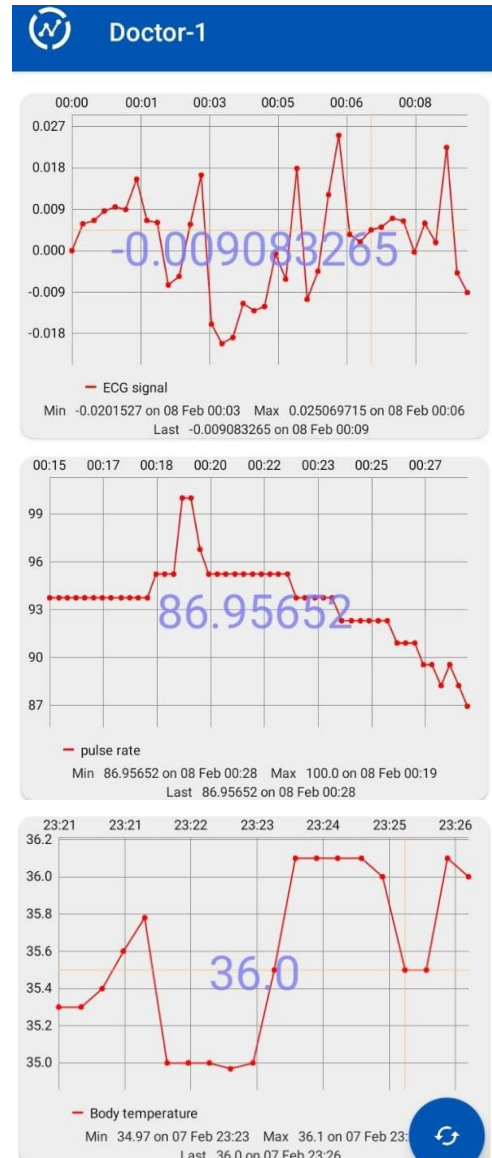


Fig. 17. Window of Doctor-1 at ThingView.

VI. CONCLUSION

In conclusion, the developed IoT-based e-health monitoring system for rural communities in Bangladesh, which integrates an ESP32 microcontroller with sensors such as the SEN-11574 pulse sensor, AD8232 ECG sensor, and LM35 temperature sensor, along with the ThingSpeak server,

represents a significant advancement in healthcare technology. This system facilitates real-time monitoring of vital health parameters, offering a comprehensive overview of patients' health statuses and enabling early detection of potential health issues. The ESP32 microcontroller, with its built-in Wi-Fi and Bluetooth capabilities, serves as the backbone of the system, ensuring seamless data collection and transmission to the ThingSpeak server. The pulse sensor measures heart rate by detecting changes in light intensity caused by blood flow, while the ECG sensor monitors the heart's electrical activity to identify potential cardiac issues. The LM35 temperature sensor provides accurate body temperature measurements. These sensors, combined with the processing and connectivity capabilities of the ESP32, allow for efficient data acquisition and real-time transmission to the ThingSpeak server, which facilitates storage, analysis, and visualization of the data. This system enables healthcare professionals to remotely access and analyze patients' health data, bridging the gap between rural and urban healthcare services. It reduces the need for frequent hospital visits and ensures timely medical interventions, particularly in resource-constrained rural areas. The affordability, non-invasive nature, and ease of implementation of this system make it an ideal solution for improving healthcare delivery in rural communities in Bangladesh. Overall, this work demonstrates the transformative potential of IoT in advancing telemedicine and improving healthcare accessibility for underserved populations.

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